

Mehdi Elahi

SoC Performance Researcher | Hardware Verification & System Optimization

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Summary

Ph.D. researcher in **Computer Engineering and Information Technology** specializing in **hardware/software co-design, low-level software engineering**, and **latency-critical edge-AI systems**. Hands-on **C, C++, and Python** developer across the full stack — from RTL and microarchitectural simulation of **multicore and MPSoC** systems to deployed production runtimes. Architected the multi-node **CREO Edge Testbed Network** and shipped **SentinelEdge**, a real-time inference system on **NVIDIA Jetson AGX Orin** at **0.044 ms** latency, published in *ACM TECS* 2026. Fluent in **Linux**, hardware diagnostics, and building systems-level tools; seeking a **2026–2027 internship** in hardware/software co-design or design Verification Roles.

Skills

Systems	Low-Level Software Engineering			Hardware/Software Co-Design				Computer Architecture	
	Data Structures & Algorithms		Performance Optimization		Embedded Systems		Multicore & MPSoC		
	Memory Systems (DRAM)								
Languages	SystemVerilog	Verilog	VHDL	Python	C	C++	Bash		
Infrastructure	Linux	Git	HPC Simulation Workflows				Isolated-Subnet Testbed Networks		
	Benchmark / Workload Development			Hardware Diagnostics & Log Analysis					
ML / AI	PyTorch	scikit-learn	NumPy	Matplotlib					
Simulation	Aldec Riviera Pro	Xilinx ISE	Vivado	GEM5	DRAMsim3	Ramulator	HSpice	Xyce	
Hardware	NVIDIA Jetson AGX Orin		NVIDIA Jetson Orin Nano			Xilinx FPGA	SPI / UART / I2C		
OS	Ubuntu	Debian	CentOS	Windows					

Technical Experience

1- SystemVerilog RTL Verification

Hands-On Verification & Low-Level Testing

2025 – Present
Greensboro, NC

- Designed **SystemVerilog** verification environments for foundational RTL components including **D Flip-Flop** and **FIFO**, validating reset behavior, timing edge cases, and depth-boundary conditions through directed and randomized test scenarios.
- Built protocol-level verification environments for **APB, AXI, and AHB** bus-memory interfaces, covering read/write transactions, wait-state handling, and back-pressure conditions across multiple master-slave configurations.
- Developed object-oriented testbenches leveraging **SystemVerilog interfaces**, inter-process communication (**IPC**) via mailboxes and semaphores, and **constrained randomization** to maximize stimulus coverage while maintaining reproducibility.
- Implemented verification environments for **SPI, UART, and I2C** serial protocols, validating functional correctness across full-duplex, half-duplex, and multi-device configurations under varied clock and baud-rate settings.
- Practiced **simulation-driven debugging** workflows — analyzing waveforms, isolating protocol violations, and iterating on DUT behavior — building hands-on intuition for pre-silicon functional validation across heterogeneous design-under-test environments.

2- Graduate Research Assistant

CREO Lab, NC A&T State University

Jan. 2024 – Present
Greensboro, NC

- Simulation & Evaluation Infrastructure.** Built simulation and evaluation infrastructure for **multicore and MPSoC** systems in **C/C++ and Python** — including benchmark development, workload analysis, and validation of hardware behavior against expected results — using **GEM5, DRAMsim3, and Ramulator** on **Linux** compute nodes.
- Low-Latency Edge Inference (SentinelEdge).** Built and deployed an attention-based deep-learning inference runtime for real-time anomaly detection, achieving up to **0.90 F1-score** at **0.044 ms** average inference latency on **NVIDIA Jetson AGX Orin**; implemented in Python/PyTorch with latency-critical profiling and optimization.

Published in **ACM TECS**, Jan. 2026.

3- Edge-Layer Network / Infrastructure Development

CREO Lab — NC A&T State University

Spring 2026
Greensboro, NC

- Architected and deployed the **CREO Edge Testbed Network** with **isolated-subnet infrastructure**, providing reproducible, secure **Linux** compute for experimentation across multiple embedded-AI nodes (NVIDIA Jetson, Xilinx FPGA).
- Standardized node provisioning and configuration so researchers could iterate faster and run experiments more reliably.

4- Workshop Instructor & Teaching Assistant

CREO Lab / VICEROY Scholars; Computer System Security Course

Sep. 2025 – Mar. 2026
Greensboro, NC

- Designed and delivered a 7-session hardware-verification curriculum (UVM methodology, hands-on FPGA projects), demonstrating strong **communication and collaboration** with student engineers.
- Presented a live adversarial hardware demo at **HOST 2026** (IEEE Intl. Symp. on Hardware Oriented Security & Trust), communicating low-level system behavior to a technical audience.

5- Digital Design Engineer

Digital Design and Hardware Development

Feb. 2022 – Nov. 2023

- Implemented digital-hardware projects (FFT-based frequency estimation, FIR/IIR filters, pipelined architectures) and designed **Network-on-Chip router architectures**, applying **algorithm design** and hardware/software co-design.

Education

1- North Carolina A&T State University

Ph.D. in Information Technology (Computer Engineering focus)

Jan. 2024 – Present
Greensboro, NC

- **Specialization:** Computer Architecture, Hardware Security, Multiprocessor SoC **GPA:** 4.0 / 4.0
- **Lab:** Center of Excellence in Cybersecurity Research, Education and Outreach (CREO)

2- Iran University of Science and Technology

M.Sc. in Electrical Engineering (Digital Design)

2015 – 2018
Tehran, Iran

- **Specialization:** Network-on-Chip, Digital RTL Design, FPGA Architecture **GPA:** 4.0 / 4.0

Event Participations

HOST 2026	IEEE Intl. Symposium on Hardware Oriented Security and Trust, Washington, D.C. — Live Hardware Demo Presenter
ISQED 2025	International Symposium on Quality Electronic Design, San Francisco, CA — Paper Presenter
MWSCAS 2025	IEEE Midwest Symposium on Circuits and Systems, Lansing, MI — Paper Presenter
ESWEEK 2024	Embedded Systems Week, Raleigh, NC
ISVLSI 2024	IEEE Computer Society Annual Symposium on VLSI — Paper Presenter

Selected Publications

[1] [M. Elahi](#), M. R. Elshamy, A.-H. Badawy, and A. Patooghy, “SentinelEdge: An Attention-Based Defense for Real-Time Mitigation of Adversarial Thermal Manipulations in System-on-Chips,” *ACM Trans. Embed. Comput. Syst.*, Mar. 2026, doi:10.1145/3802543.

[2] [M. Elahi](#), A. Badawy, and A. Patooghy, “DriverHammer: Hardware-Assisted RowHammer via Covert Wordline Driver Amplification in DRAM,” in *Proc. 2026 IEEE/ACM Intl. Conf. on Computer-Aided Design (ICCAD)*, 2026 (Submitted).

[3] [M. Elahi](#), M. R. Elshamy, A.-H. Badawy, and A. Patooghy, “iThermTroj: Exploiting Intermittent Thermal Trojans in Multi-Processor System-on-Chips,” in *Proc. 2025 IEEE 68th Intl. Midwest Symp. on Circuits and Systems (MWSCAS)*, 2025, pp. 912–916, doi:10.1109/MWSCAS53549.2025.11244435.

[4] [M. Elahi](#) and H. S. Shahhosseini, “LTD-Router: Low Latency Traffic Distributed FPGA Based Router Architecture Using Dedicated Paths,” in *Proc. Iranian Conf. on Electrical Engineering (ICEE)*, Mashhad, Iran, 2018, pp. 1601–1606, doi:10.1109/ICEE.2018.8472707.

Full list of 12 publications available on Google Scholar.